TODO: Jag har försökt lägga fram kärnan i vad jag tror kommer att bli uppsatsen, men många delar är ännu inte färdiga. Jag har försökt lämna färgade kommentarer här och var så att du kan se hur jag har tänkt samt vilka frågeställningar jag har.
Utöver det är jag inte helt säker på mina val av felmodeller (convolution) och stickprovsstorlekar. Jag har skrivit min kod i R som en pipeline, så dokumentet kan enkelt uppdatera befintliga grafer och plottar om du tycker att jag bör använda andra värden.
Jag är också osäker på hur mycket av materialet som bör tas med i uppsatsen i dess nuvarande form. I dagsläget har jag 64 olika simuleringar, vilket är en kombination av felmodell, stickprovsstorlek och dekonvolutionsmetod.

To be determined)

Emilio Otero Johansson

April 23, 2026

Abstract

TODO: Will be done at the end

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1 Introduktion

2 Introduction

Accurate knowledge of population structure is fundamental in wildlife ecology, particularly for long-lived species where survival and reproduction vary strongly with age. In Scandinavian brown bear populations, age composition plays a central role in understanding population growth, reproductive potential, and long-term sustainability. For this reason, monitoring programs, such as the Swedish Environmental Protection Agency?s national wildlife monitoring scheme, regularly collect biological data to estimate population size and structure, including age distributions of bears **nrm˙bear˙monitoring**.

A central challenge in such monitoring efforts is that age is rarely observed directly. Instead, age is typically inferred using indirect biological indicators such as tooth cementum layers or other morphological measurements. These procedures are inherently subject to measurement uncertainty and classification error. As a result, the observed age distribution does not perfectly reflect the true underlying population structure. This discrepancy can introduce bias into demographic estimates and may propagate into downstream analyses such as population forecasting and management decisions.

Recent methodological developments have therefore focused on statistical correction techniques that explicitly account for measurement error in age estimation. One such approach is based on deconvolution methods, where the observed data are modeled as a noisy transformation of an unobserved latent distribution. In this framework, the observed age distribution is expressed as a convolution between the true age distribution and a measurement error mechanism. Recovering the latent distribution then becomes an inverse problem, which is often ill-posed and sensitive to noise in the data.

A recent development in this area is the work by Nakamura et al. Nakamura et al. 2023, who propose an age estimation method for brown bears based on blood DNA methylation levels. This approach provides a molecular alternative to traditional age determination techniques, which are often based on tooth wear or cementum annuli. While such methods can improve individual-level age estimation, they still produce observations that are subject to measurement error and uncertainty, which propagates into population-level age structure estimates.

In general, any age estimation method introduces a stochastic relationship between the true age and the observed measurement. This motivates statistical approaches that explicitly model this uncertainty and attempt to reconstruct the latent age distribution from noisy observations. Such approaches can be formulated as inverse problems, where the observed data are viewed as a convolution of the true age distribution and a measurement

error mechanism. The effectiveness of these methods depends on both the severity of measurement error and the amount of available data.

The present thesis investigates this inverse problem in a controlled simulation setting. The true population age structure is generated using a Leslie matrix model, which provides a biologically grounded age-dependent population dynamic. Measurement error is then introduced through a probabilistic misclassification mechanism, producing an observed age distribution that deviates from the latent structure.

The main objective of this study is to evaluate how well the latent age distribution can be recovered using a deconvolution approach based on the Expectation-Maximization (EM) algorithm. In particular, the performance of the method is assessed under varying levels of measurement error and sample sizes. The quality of the estimated distributions is evaluated using total variation distance, while the practical implications of estimation error are further examined through population forecasting using the Leslie matrix model.

By combining a structured demographic model with a controlled measurement error framework, this study aims to provide insight into the conditions under which deconvolution methods can meaningfully improve inference about population age structure, and when such corrections may or may not translate into improved ecological forecasts.

3 Methodology

3.1 Deconvolution

Measurement error in statistical models can be described as the discrepancy between a latent (true) variable and its observed counterpart. A common way to formalize this is through a convolution structure, where the observed variable is a noisy transformation of the true signal.

$$Y = Z + \epsilon.$$

Here, Z denotes the latent random variable, ϵ denotes measurement error, and Y denotes the observed random variable. The distribution of Y is then the convolution of the distributions of Z and ϵ.

This formulation extends naturally to multivariate settings, where both the latent and observed quantities are represented as random vectors.

$$\mathbf{Y} = \mathbf{Z} + \epsilon.$$

In many applications, including the present one, the variable of interest is discrete. In such cases, the additive error formulation is replaced by a discrete measurement error mechanism, which will be specified in the following sections.

In a deconvolution problem, the goal is to recover the distribution of the latent variable based on observations of the corresponding noisy measurements and under the assumption that the distribution of the error term is fully known. Deconvolution problems are typically ill-posed in the sense that small perturbations in the observed data can lead to large changes in the estimated distribution (e.g. due to sampling variation).

Latent distribution In this study, the true age structure of the bear population is represented by the parameter vector

$$\boldsymbol{\pi} = (\pi_1, \pi_2, \ldots, \pi_k),$$

where π_j denotes the probability that an individual belongs to age class j. In particular, π_1 corresponds to the age group 0–1 years, π_2 to 1–2 years, and so on. The discretization of age into fixed classes is also consistent with population structure is often analyzed in biologically meaningful age intervals, where demographic processes such as survival and reproduction are age-class dependent as seen in Norwegian Institute for Nature Research (NINA) n.d.

Let n denote the number of independent individuals in the sample. For each individual $r = 1, \ldots, n$, let $Z_r \in \{1, \ldots, k\}$ denote the true age class. We assume that the latent age classes are independent and identically distributed with

$$P(Z_r = j) = \pi_j.$$

The vector of observed class counts is then defined as

$$\boldsymbol{Z} = \left(Z_1^{(c)}, Z_2^{(c)}, \ldots, Z_k^{(c)}\right),$$

where

$$Z_j^{(c)} = \sum_{r=1}^{n} \mathbf{1}(Z_r = j)$$

denotes the number of individuals in age class j. It follows that

$$\boldsymbol{Z} \sim \text{Multinomial}(n, \boldsymbol{\pi}).$$

Measurement error As mentioned in section 3.1 the measurement error is assumed to be fully known. In this simulation study, we assume a normal distribution for the error term as a simple and tractable way to control the variability in the observations and to introduce systematic measurement error. While this choice may not exactly reflect the error structure described in Nakamura et al. 2023, the aim of this thesis is to investigate the recovery of the latent distribution under a simplified but comparable error setting.

Additionally, we assume that the measurement error depends on the true age class Z_r. Specifically, we model

$$\epsilon_r \mid Z_r = j \sim \mathcal{N}(0, \sigma^2).$$

Applying the measurement error directly at the individual level would yield a continuous latent measurement $Z_r + \epsilon_r$, which is incompatible with the discrete age-class structure of the model. We therefore define a discrete observation mechanism induced by this continuous error model by discretizing the conditional distribution of $\epsilon_r \mid Z_r = j$ over the k predefined age classes.

Let the probabilities M_{ij} denote the probability of misclassifying a bear of true age j with age i:

$$M_{ij} = P(Y_r = i \mid Z_r = j)$$

Where $Y_r \in \{1, \ldots, k\}$ denotes the observed (possibly misclassified) age class corresponding to individual r such that:

$$\sum_{i=1}^{k} M_{ij} = 1$$

This leads to the construction of a square missclassification matrix M with k rows and columns. Representing observed age class i and true age j respectively.

Observed data Given the misclassification mechanism described above, each individual r with true age class j is observed in age class i with probability M_{ij}. Consequently, the observed counts $\mathbf{Y} = (Y_1, \ldots, Y_k)$ follow a multinomial distribution with probabilities given by the transformed latent distribution:

$$\mathbf{Y} \sim \text{Multinomial}(n, M\boldsymbol{\pi}). \tag{1}$$

Naive estimator The objective of this thesis is to investigate how well the true age distribution $\boldsymbol{\pi}$ can be recovered from misclassified observations. In the absence of measurement error, a natural estimator of $\boldsymbol{\pi}$ is given by the empirical proportions

$$\hat{\boldsymbol{\pi}} = \frac{\mathbf{Z}}{n} \tag{2}$$

which satisfies $\mathbb{E}[\hat{\boldsymbol{\pi}}] = \boldsymbol{\pi}$.

However, in the presence of measurement error, we observe $\mathbf{Y}$ instead of $\mathbf{Z}$, and the naive estimator $\mathbf{Y}/n$ is generally biased for $\boldsymbol{\pi}$. This motivates the deconvolution approach considered in this thesis.

3.2 Estimation methods

In this thesis, we consider two methods for estimating the latent age structure $\boldsymbol{\pi}$: a method-of-moments (MoM) estimator and a maximum likelihood estimator based on the EM algorithm.

3.2.1 Method of Moments

The method-of-moments assumes that population moments, such as $\mathbb{E}[\mathbf{Z}]$, can be expressed as functions of the unknown parameter $\boldsymbol{\pi}$. Parameter estimates are obtained by equating these theoretical moments with their empirical counterparts Alm and Britton 2008, p. 275.

From equation (1), the first moment of the observed data is given by

$$\mathbb{E}[\boldsymbol{Y}] = nM\boldsymbol{\pi}. \tag{3}$$

The corresponding empirical moment is given by the observed proportions $\boldsymbol{Y}/n$ see equation 2. Equating theoretical and empirical moments yields

$$M\boldsymbol{\pi} = \frac{\boldsymbol{Y}}{n}. \tag{4}$$

Solving for $\boldsymbol{\pi}$ gives the MoM estimator

$$\hat{\boldsymbol{\pi}} = M^{-1}\frac{\boldsymbol{Y}}{n}, \tag{5}$$

provided that M is invertible.

3.2.2 Maximum-likelihood method ML

The ML method method for parameter estiamte consist of arg maxing the log likelihood function for a parameter. Which in our current multinomial case coinced with the MoM thus leading to the same problem of matrix inversion (negative estiamtes , probabilities not summing to 1and so on). We therefore present the em-alg presented bellow

Expextacion Maximization algorithm EM-algorithm The EM algorithm is a general-purpose method for computing maximum likelihood estimates in the presence of incomplete or latent data (McLachlan and Krishnan 2008). Let $\mathbf{Y}$ denote a discrete random vector corresponding to the observed (incomplete) data, and $\mathbf{X}$ a discrete random vector representing the complete data. If $\mathbf{X}$ were fully observable, we could directly compute the complete-data log-likelihood function $\ell_c(\boldsymbol{\pi})$ based on a realization x.

However, in practice we only observe $y \in \mathcal{Y}$, the sample space of $\mathbf{Y}$. The complete-data sample space $\mathcal{X}$ is related to $\mathcal{Y}$ through a many-to-one mapping, i.e., a surjection from $\mathcal{X}$ to $\mathcal{Y}$, where multiple elements $x \in \mathcal{X}$ correspond to the same observed value y.

The probability mass function of $\mathbf{Y}$ is therefore given by

$$p_{\mathbf{Y}}(y; \boldsymbol{\pi}) = \sum_{x \in \mathcal{X}(y)} p_{\boldsymbol{X}}(x; \boldsymbol{\pi}), \tag{6}$$

where $\mathcal{X}(y) = \{x \in \mathcal{X} : h(x) = y\}$.

Since the complete data $\mathbf{X}$ is not observable, the complete-data log-likelihood $\ell_c(\boldsymbol{\pi})$ cannot be evaluated directly. Instead, the EM algorithm considers its conditional expectation given the observed data y and a current parameter estimate $\boldsymbol{\pi}^{(t)}$ McLachlan and Krishnan 2008:

$$Q(\pi \mid \boldsymbol{\pi}^{(t)}) = \mathbb{E}_{\pi^{(t)}}\left[\ell_c(\boldsymbol{\pi}) \mid Y = y\right] \tag{7}$$

which corresponds to the Expectation step (E-step) of the EM algorithm. In the Maximization step (M-step), the parameter is updated by maximizing this function:

$$\pi^{(t+1)} = \arg\max_{\pi} Q(\pi \mid \pi^{(t)}). \tag{8}$$

The process is then iterated upon until $\ell(\pi^{t+1}) - \ell(\pi^t) <= k$ for some arbitrary small k value, which is then interpreted as a local/global maxima.

Monotonicity . A key property for the algorithm to work is for the likelihood to be monotonically increasing for each iteration. $L(\theta^{t+1}) \geq L(\theta^t))$ which is discussed in Section 3.2 of McLachlan and Krishnan 2008.

Applying EM-alg In principle, the complete data could be represented by individual-level pairs $X = (Z, Y)$, where Z denotes the true age class and Y the observed age class for each individual. However, in the present multinomial setting it is more convenient to work with N_{ij}, where N_{ij} denotes the number of individuals with true age class j that are observed in age class i.

The PMF for X is then:

$$p_X(x = (z_j, y_i); \pi_j) = \pi_j M_{ij} \tag{9}$$

where M is the misclasification matrix defined in 3.1 and for a vector of n independent observations $x_v = (z_1, y_1), (z_2, y_2), ..., (z_n, y_n)$ we obtain

$$p_X(x_v; \boldsymbol{\pi}) = \prod_i \prod_j (\pi_j M_{ij})^{n_{ij}} \tag{10}$$

where $n_{ij} = \sum_{n=1} 1\{Z_r = j, Y_r = i\}$ is a random variable with the counts that were observed as i given they were j.

The log-likelihood is then:

$$\ell_c(\boldsymbol{\pi}) = \sum_j \sum_i n_{ij} \log(p_j M_{ij}) = \sum_j \sum_i n_{ij} \log M_{ij} + \sum_j \sum_i n_{ij} \log \pi_j \quad (11)$$

In accordance with equation (7) we take the conditional expectation on n_{ij} with an initial parameter $\boldsymbol{\pi}^{(\mathbf{0})}$ conditioned on Y

$$\mathbb{E}_{\boldsymbol{\pi}^{(0)}}\left[\ell_c(\boldsymbol{\pi}) \mid \mathbf{Y} = \mathbf{y}\right] = \mathbf{E}_{\pi^{(0)}}\left[\sum_j \sum_i n_{ij} \log(\pi_j M_{ij}) \mid \mathbf{Y} = \mathbf{y}\right] = \sum_i \sum_j \mathbf{E}_{\boldsymbol{\pi}^{(0)}}\left[n_{ij} \mid \boldsymbol{Y} = \boldsymbol{y}\right] \log(\pi_j M_{ij}) \quad (12)$$

The conditional expectation $E_{\boldsymbol{\pi}^{(\mathbf{0})}}[n_{ij} \mid \mathbf{Y} = \mathbf{y}]$ is the expected number of bears with true age class j that are observed in age class i, given the observed data $\mathbf{Y} = \mathbf{y}$. Note that this depends on the assumed latent age distribution $\boldsymbol{\pi}^{(\mathbf{0})}$.

Let $Y_i^{(c)}$ denote the number of observed individuals in age class i:

$$Y_i^{(c)} = \sum_{r=1}^{n} \mathbf{1}\{Y_r = i\} \quad (13)$$

The expected number of individuals with true age class j, among those observed in age class i, is given by the number of observed individuals $Y_i^{(c)}$ multiplied by the probability of originating from true age class j given observation in class i, $P(Z = j \mid Y = i, \boldsymbol{\pi}^{(0)})$. This probability is proportional to the product of the misclassification probability M_{ij} and the assumed true proportion $\pi_j^{(0)}$, and is normalized by the total probability of being observed in class i from any true age class.

$$P(Z = j \mid Y = i, \boldsymbol{\pi}^{(0)}) = \frac{\pi_j^{(0)} M_{ij}}{\sum_{a=1}^{k} \pi_a^{(0)} M_{ia}}$$

Thus:

$$\mathbb{E}[n_{ij} \mid Y = y] = n_j \cdot \frac{p_i^{(0)} M_{ij}}{\sum_{i'} p_{i'}^{(0)} M_{i'j}} \quad (14)$$

Maximization-step

Maximization-step We now choose $\boldsymbol{\pi}^{(t+1)}$ such that, in accordance with equation (8), the conditional expected log-likelihood is maximized. From equation (11), we note that only the terms involving $\boldsymbol{\pi}$ depend on the parameter of interest, since the remaining terms do not depend on $\boldsymbol{\pi}$ and can therefore be ignored in the maximization.

Thus, maximizing the conditional log-likelihood is equivalent to maximizing

$$\sum_j \sum_i \mathbb{E}[n_{ij} \mid \mathbf{Y} = \mathbf{y}] \log \pi_j.$$

3.3 Matrix population models

To obtain a suitable idea of the population age distribution over time we use a population matrix model Caswell 2001. The age class $j = 1, 2, ...k$ is the discretization of the age z to age-class j with $j - 1 \leq z \leq j$. The number of individuals for each age class j is then given by $n_1, n_2, ..., n_k$.

Following Caswell 2001, Eq. 2.1 the number of n_1 individuals at time $t + 1$ can be projected with

$$n_1(t + 1) = f_1 n_1(t) + f_2 n_2(t) + ... + f_k n_k(t)$$

Here f_j denotes the reproductive rate per-capita. That is the average number of offspring an individual of age class j has. For $j > 1$ the number of individuals in an age class n_j at time $t+1$ is simply given by the survival probability of previous age class $(s_{j-1})(n_{j-1})$ at the previous timestep t. Together we can write the equation in matrix form for all n_j at time $t + 1$.

$$\begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_k \end{pmatrix} (t+1) = \begin{pmatrix} f_1 & f_2 & f_3 & \cdots & f_k \\ s_1 & 0 & 0 & \cdots & 0 \\ 0 & s_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & s_{k-1} & 0 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \\ \vdots \\ \vdots \\ n_k \end{pmatrix} (t)$$

3.4 Computing age distribution from a Leslie matrix

The stable age distribution of a population can be obtained from the Leslie matrix $\mathbf{L}$ by computing its dominant eigenvalue λ_1 and the corresponding right eigenvector $\mathbf{w}$. This eigenvector gives the relative proportions of individuals in each age class at equilibrium, and can be normalized to represent a probability vector for k distinct age classes.

The existence of a real, non-negative eigenvalue λ_1 that is greater than or equal in magnitude to all other eigenvalues λ_i of a non-negative, primitive matrix $\mathbf{A}$ is guaranteed by the Perron-Frobenius theorem Caswell 2001.

If $\mathbf{L}$ is primitive, then as $t \to \infty$, the population vector $\mathbf{n}_t$ satisfies

$$\frac{\mathbf{n}_t}{\sum_j n_{j,t}} \to \mathbf{w},$$

where $\mathbf{w}$ is the normalized right eigenvector corresponding to λ_1. This represents the **stable age distribution** of the population and the λ_1 is the growth rate for a stable population.

3.5 Evaluation method

Total Varation (TV) To quantify the difference between two probability distributions P and Q over the same finite sample space $\mathcal{X}$, we use the *Total Variation distance (TV)*:

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)|.$$

The TV distance ranges from 0 (distributions are identical) to 1 (distributions have no overlap). Intuitively, it represents the maximum fraction of probability mass that must be reallocated to transform P into Q or vice versa. In this thesis, we use TV distance to quantify and distinguish the effects of sampling variability, convolution, and deconvolution on our estimates of the latent distribution (true age distribution).

[TODO: Does this need a source reference?]

4 Simulation Study

4.1 Simulated Age Distributions

We consider a population structured into $k = 20$ age classes, consistent with the discretisation used in Norwegian Institute for Nature Research (NINA) n.d. Based on this structure, we construct a 20×20 Leslie matrix $\mathbf{L}$ as described in section 3.3. Each age class is defined as a one-year interval, where the age-class index $j \in \{1, \ldots, 20\}$ corresponds to the interval $[j-1, j)$ for $j = 1, \ldots, 19$, while the final class $j = 20$ represents individuals aged 19 years and older.

We define the fertility contribution of individuals in age class j as

$$f_j = s_j \cdot p_{\mathrm{coy},j} \cdot \mathrm{litter}_j, \tag{15}$$

where:

- s_j is the survival probability from age j to $j + 1$,
- $p_{\mathrm{coy},j}$ is the probability of conceiving at age j,
- litter_j is the expected litter size at age j.

The survival and reproduction parameters were approximated by visual inspection of published graphs Norwegian Institute for Nature Research (NINA) n.d. The extracted values used in the construction of $\mathbf{L}$ are provided in Appendix 7.1.

The stable age distribution, shown in Figure 1, is computed as the normalized dominant right eigenvector of $\mathbf{L}$ using the method described in Section 3.4. This distribution is denoted by $\boldsymbol{\pi}$ and serves as the latent age distribution used throughout the simulation study.

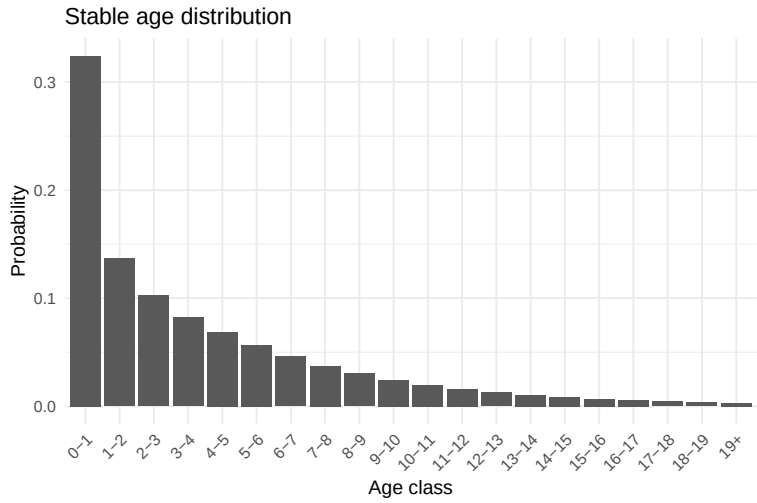


Figure 1: Latent age distribution $\boldsymbol{\pi}$ obtained from the normalized dominant eigenvector of the Leslie matrix $\mathbf{L}$.

4.2 Error terms

In this section, we define the discretization procedure used to construct the misclassification matrix M described in Section 3.1. The chosen error model is designed to approximately reflect the error pattern shown in Figure 5(a) in Nakamura et al. 2023. This choice was motivated by a visual inspection of several plausible discretization strategies, in particular different treatments of boundary and out-of-range values.

These included truncation of negative values at zero, absorption into the youngest age class (0–1 years), truncation of such values followed by renormalization of probabilities, and transformation via absolute values prior to binning.

Error model of choice For each individual with true age class $Z_r = j$, we assume an additive Gaussian measurement error as specified in 3.1

$$\epsilon_r \mid Z_r = j \sim \mathcal{N}(0, \sigma^2).$$

In the simulation study, we fix $\sigma = 4$, which was chosen to approximate the spread of the measurement error observed in Figure 5(a) of Nakamura et al. 2023.

The observed value is then implicitly given by $j - 0.5 + \epsilon_r$, and is mapped to discrete age classes by binning into intervals $[i-1, i)$.

To account for the lower boundary, we condition on the event that the perturbed value lies within the support of the age scale, resulting in a truncated distribution. The probability of observing class $i = 1, \ldots, 19$ is there-

fore

$$M_{ij} = \frac{\Phi\left(\frac{i-(j-0.5)}{\sigma}\right) - \Phi\left(\frac{i-1-(j-0.5)}{\sigma}\right)}{1 - \Phi\left(\frac{0-(j-0.5)}{\sigma}\right)}.$$

For the upper boundary, the last age class absorbs all remaining probability mass:

$$M_{20,j} = 1 - \sum_{i=1}^{19} M_{ij}.$$

This ensures that $\sum_{i=1}^{20} M_{ij} = 1$.

Applying the misclassification matrix M to the latent age distribution $\boldsymbol{\pi}$ yields the observed age distribution $\boldsymbol{\pi}^{\text{obs}} = M\boldsymbol{\pi}$, as defined in Section 3.1.

$$\boldsymbol{\pi}^{\text{obs}} = M\boldsymbol{\pi}.$$

The resulting distribution corresponding to $\sigma = 4$ is shown in Figure 2.

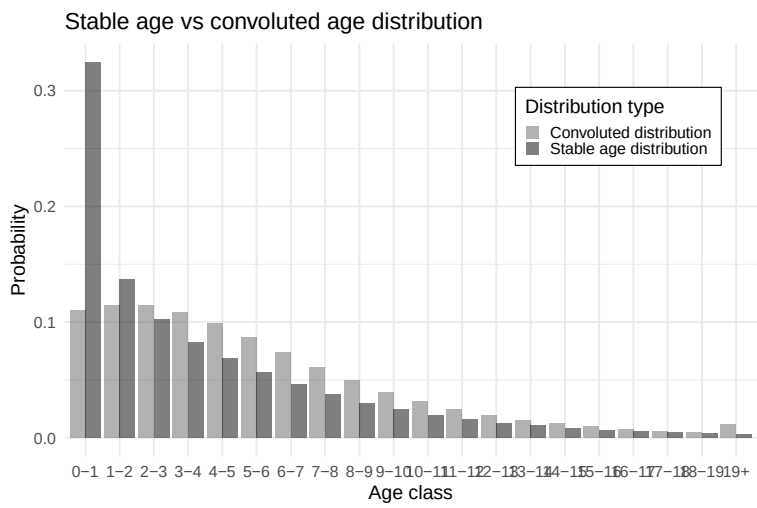


Figure 2: Stable age distribution $\boldsymbol{\pi}$ and the corresponding observed age distribution $M\boldsymbol{\pi}$ under the measurement error model $\epsilon_r \mid Z_r \sim \mathcal{N}(0, 4^2)$.

4.3 Simulation design

This section describes the experimental design used to evaluate the performance of the proposed deconvolution methods under controlled conditions.

General setup The latent age distribution $\boldsymbol{\pi}$ is fixed across all simulations and corresponds to the stable age distribution of a Leslie matrix model with $k = 20$ age classes (see Section 3.4). Measurement error is

introduced through a misclassification matrix M, constructed from the discretized Gaussian error model described in Section 4.2, with fixed standard deviation $\sigma = 4$.

This setup ensures that all variation in the results arises from sampling variability and estimation error, rather than changes in the underlying population structure.

Sample sizes To investigate how estimation performance depends on sample size, we consider:

$$n \in \{50, 100, 250, 1000\}.$$

Estimators We compare three estimators of the latent distribution:

- the naive empirical estimator $\hat{\boldsymbol{\pi}} = \mathbf{Y}/n$,
- the method-of-moments (MoM) estimator,
- the EM-based maximum likelihood estimator (see Section 3.2.2).

Simulation repetitions For each combination of sample size and estimator, the simulation is repeated R times (typically $R = 1000$) to obtain Monte Carlo estimates of bias, variability, and overall performance.

EM algorithm settings For the EM-based estimator, convergence is declared when the log-likelihood improvement satisfies

$$\ell(\boldsymbol{\pi}^{(t+1)}) - \ell(\boldsymbol{\pi}^{(t)}) < \epsilon,$$

with $\epsilon > 0$ small (typically 10^{-4}), or when a maximum number of iterations (100) is reached.

4.4 Simulation procedure

To evaluate the performance of the deconvolution methods, we conduct a Monte Carlo simulation study. Each replication follows the steps below.

Step 1: Generate latent population For a given sample size n, we simulate latent age classes independently from the true distribution:

$$Z_r \sim \text{Categorical}(\boldsymbol{\pi}), \quad r = 1, \ldots, n.$$

This yields the latent count vector

$$\mathbf{Z}^{(c)} \sim \text{Multinomial}(n, \boldsymbol{\pi}).$$

Step 2: Introduce measurement error Each latent observation is passed through the misclassification mechanism defined by M. Conditional on $\mathbf{Z}^{(c)}$, the observed counts are generated as

$$\mathbf{Y}^{(c)} \sim \text{Multinomial}(n, M\boldsymbol{\pi}).$$

This represents the convolution of the latent distribution with the measurement error process.

Step 3: Estimation Given the observed data $\mathbf{Y}^{(c)}$, we estimate the latent distribution using three methods:

- the naive estimator $\hat{\boldsymbol{\pi}}^{\text{naive}} = \mathbf{Y}^{(c)}/n$,
- the method-of-moments estimator,
- the EM-based maximum likelihood estimator (Section 3.2.2).

Step 4: Performance evaluation For each estimator, we compute the total variation (TV) distance to the true distribution:

$$\text{TV}(\hat{\boldsymbol{\pi}}, \boldsymbol{\pi}) = \frac{1}{2} \sum_{j=1}^{k} |\hat{\pi}_j - \pi_j| \,.$$

This measure is used to quantify the estimation error induced by sampling variability and measurement error.

Step 5: Forecasting analysis To assess the practical implications of estimation error, we propagate the estimated age distributions through the Leslie matrix model:

$$\mathbf{n}_{t+1} = \mathbf{L}\,\hat{\boldsymbol{\pi}}_t.$$

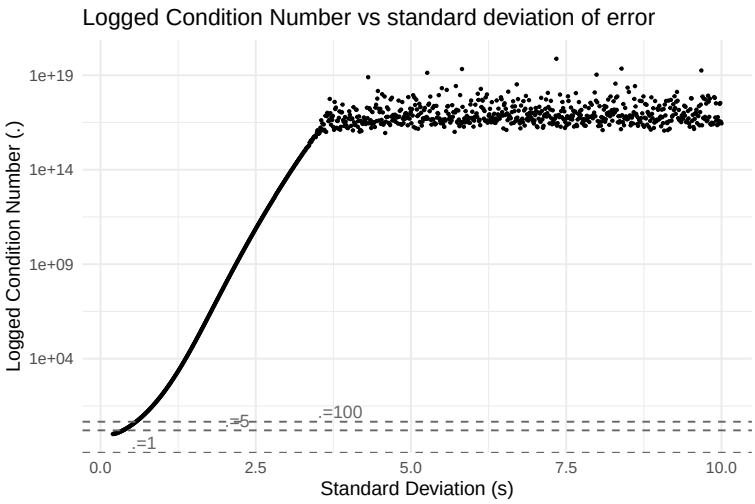
Forecasts based on estimated distributions are compared to those obtained using the true latent distribution. Differences are evaluated over multiple time steps to assess how estimation error propagates through population dynamics.

4.5 Deconvolution Methods

TODO: This section depends if i will have the time or should attempt any other method for deconvolution i leave it as it is for after feedback

Two deconvolution methods were used and compared: a thikanov regularization matrix inversion and the EM algorithm described in Section **??**. For each error model introduced in Section 4.2. The resulting estimates were evaluated using the KL-divergence, as described in Section **??**.

Matrix inversion The matrix inversion method inverts the misclassification matrix that can then be used to calculate the recovered distribution, negative estimates are set to 0 and the distribution is then normalized. While straightforward, this approach can become unstable and or ill-conditioned as σ for the error distribution increases as shown in 4.5.



EM-algorithm deconvolution We applied the EM algorithm (see Appendix **??**) using two different initialization: a uniform distribution, representing a neutral/uninformed initial belief about the latent age distribution, and the true age distribution.

The initialization based on the true age distribution was included to assess how a well-informed prior affects the resulting estimate compared to a neutral one. In particular, this allows us to evaluate the sensitivity of the EM algorithm to initialization and to investigate for which sample sizes initialization no longer constitutes a confounding factor in an ideal setting.

The algorithm was iterated until convergence, defined as

$$\ell_c(\theta^{(t+1)}) - \ell_c(\theta^{(t)}) \leq 10^{-4},$$

or until a maximum of 100 iterations was reached.

5 Results

TODO: text and polish is largely unfinished but the main results to describe are there in plots and tables

5.1 Effect of convolution and sampling

Figure **??** shows how sample size affects the total variation distance (TV distance) across all models. For small sample sizes, sampling noise significantly affects the TV distance. As the sample size increases, the TV distance gradually stabilizes across all models, suggesting that sampling noise plays a diminishing role relative to the structural effects induced by the convolution. Sampling from the true model is included as a theoretical baseline for sampling noise.

For the unbiased Models **??** and **??**, most of the variation in TV distance is driven by sampling noise, whereas for the biased Models **??** and **??**, the TV distance is primarily driven by convolution structure. Additionally, Model **??**, which is characterized by lower sd ($\sigma = 1$) compared to Model **??** ($\sigma = 4$), exhibits systematically higher total variation distance.

The median and non-parametric 95% upper quantile also exhibit a consistent decreasing pattern with increasing sample size, in line with the results observed for the mean (see Table **??**).

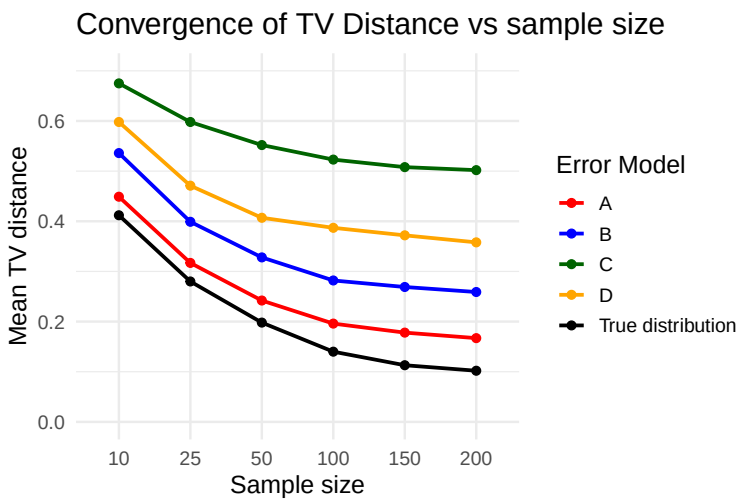


Figure 3: Mean TV distance for convoluted distributions for Model **??**–**??**, with 100 replicas per sample size. Total variation distance is computed relative to the true distribution.

5.2 Deconvolution

Figure 4 and Figure **??**

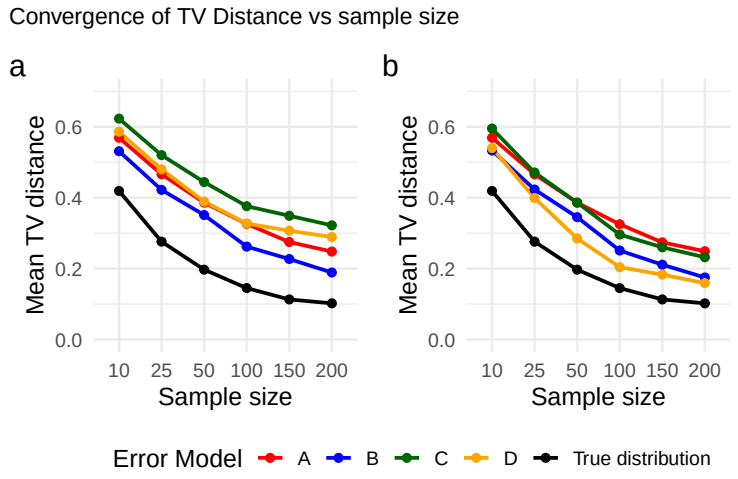


Figure 4: Mean total variation (TV) distance for deconvolved samples based on an uninformed initialization for Models **??**–**??**, using 100 replicates per sample size. The TV distance is computed relative to the true distribution. Panel (a) shows the EM algorithm with a uniform (uninformed) initialization, while panel (b) shows results with initialization using the true (informed) distribution.

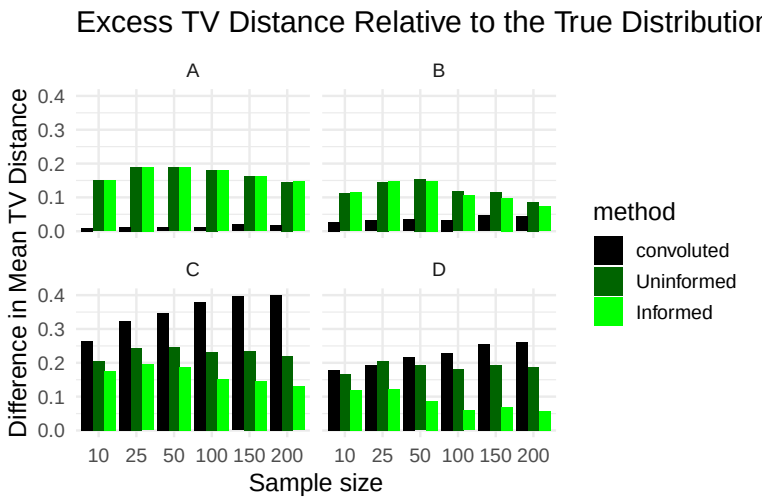


Figure 5: Mean total variation (TV) distance for convolved and deconvolved samples based on an uninformed and informed initialization, using 100 replicates per sample size. The TV distance is computed relative to the true distribution. Panel (A–D) corresponding to error Models **??**–**??**.

model	sample_size	comparison	diff	stars
C	200	I-C	-0.262	***
C	150	I-C	-0.247	***
C	100	I-C	-0.224	***
D	200	I-C	-0.190	***
D	150	I-C	-0.185	***
C	200	U-C	-0.175	***
D	100	I-C	-0.173	***
A	25	U-C	0.163	***
A	25	I-C	0.162	***
C	50	I-C	-0.161	***

Table 1: Table displaying the top 10 differences in total variation (TV) distance between convoluted and deconvoluted samples. Statistical significance is assessed using paired t-tests, where * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

TODO: Plot 6 and Plot 7 tell the main story how sample size, deconvolution vs convoluted samples affect total population forecasts. I didnt have time to describe these properly and what can be observed. TV was used more as a diagonostic tool for the deconvolution but does not say much about weather or not forecast would be good using it. E.g. if most of the TV can be explained my shifts in the probability mass of age classes that have 0 fertility this would not affect the total population forecast the coming year. The plots were made by simulating 100 samples for each combination and taking $L^n(p_{error} - p_{true})$ for $n = 1, 2, ...5$

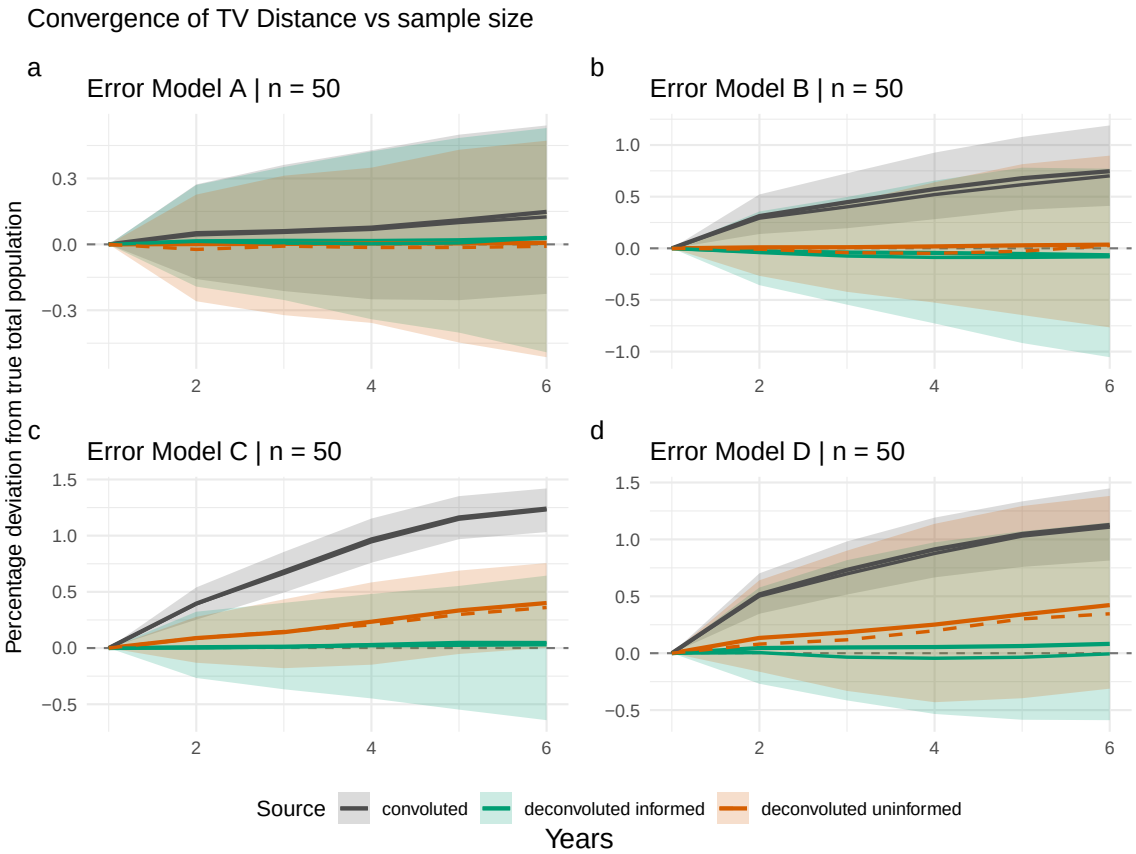

Convergence of TV Distance vs sample size
a
Error Model A | n = 50
b
Error Model B | n = 50
c
Error Model C | n = 50
d
Error Model D | n = 50
Percentage deviation from true total population
Source
convoluted
deconvoluted informed
deconvoluted uninformed
Years

Figure 6: s

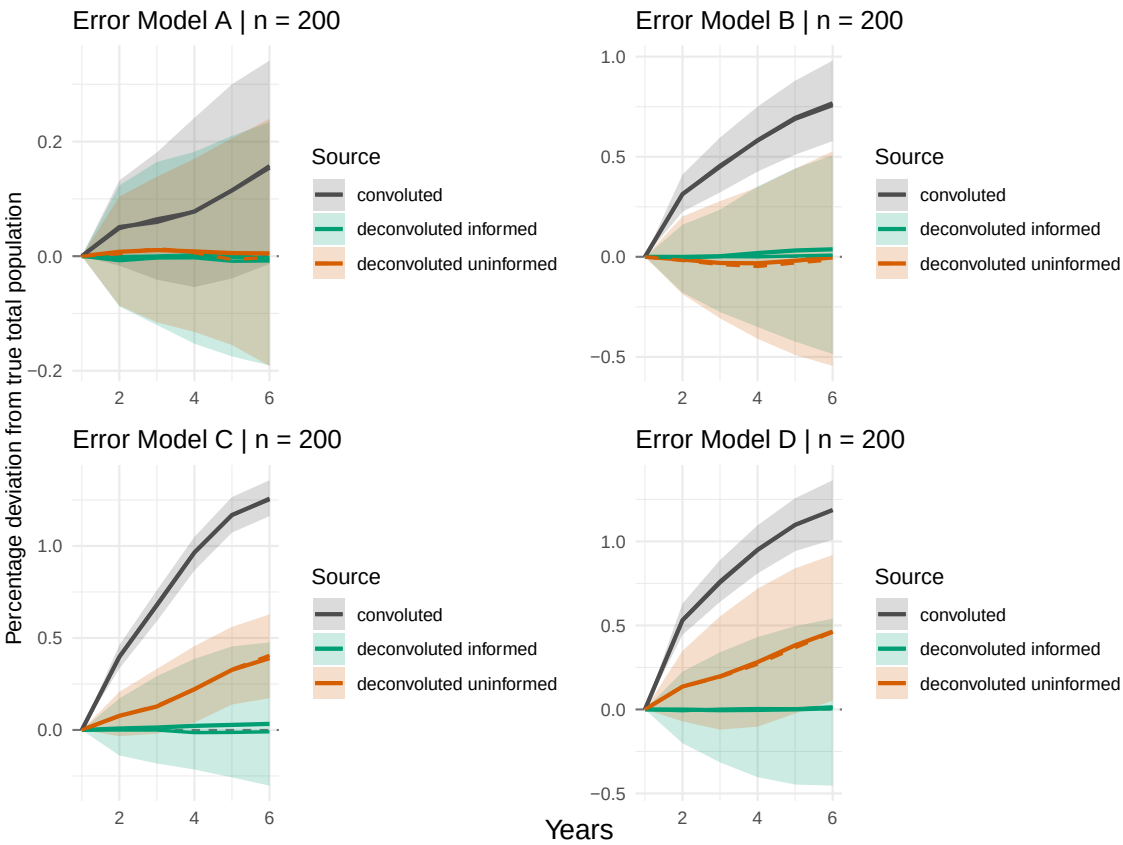


Figure 7: s

6 Analysis / Discussion

TODO:

- Under which conditions the deconvolution method is reliable and appropriate, in terms of total variation (TV), sample size, and error structure.
- Analyze how sampling variability interacts with different error models, and whether it affects deconvolution performance differently across these settings.
- Compare informed vs uninformed deconvolution approaches vs keep convoluted. Assess whether informed initialization leads to improved performance, and how this relates to convergence properties of the expected conditional log-likelihood, including the probability of convergence to different local optima. Suggest as improvement whether multiple random initializations with selection based on arg max log-likelihood, or structural parameter constraints (e.g., monotonicity constraints such as bin 1 $\geq$ bin 2 $\geq \ldots$).
- Evaluate whether total variation relates or not to forecasts. Is the variability of the forecasts more dominated by the sampling error vs ... Does the deconvolution improve forecasts relative to the convolution? In which way ? bias , variability?
- Why does the em-alg behave differently for each model? Is decon-

7 Appendix

7.1 Simulation Parameters

Table 2: Parameters used to construct the Leslie matrix.

Age	Survival	p_coy	Litter_size	F_i
1	0.50	0.00	0.00	0.00
2	0.89	0.00	0.00	0.00
3	0.95	0.00	0.00	0.00
4	0.99	0.00	0.00	0.00
5	0.98	0.15	2.12	0.31
6	0.96	0.35	2.25	0.76
7	0.96	0.55	2.40	1.27
8	0.96	0.55	2.40	1.27
9	0.96	0.55	2.40	1.27
10	0.96	0.55	2.40	1.27
11	0.96	0.55	2.40	1.27
12	0.96	0.70	2.60	1.75
13	0.96	0.70	2.60	1.75
14	0.96	0.70	2.60	1.75
15	0.96	0.70	2.60	1.75
16	0.96	0.70	2.60	1.75
17	0.96	0.58	2.75	1.53
18	0.96	0.58	2.75	1.53
19	0.96	0.58	2.75	1.53
20	0.96	0.58	2.75	1.53

7.2 Results

model	sample_size	mean	median	upper_95
True distribution	10	0.412	0.410	0.527
True distribution	25	0.280	0.280	0.367
True distribution	50	0.198	0.200	0.271
True distribution	100	0.140	0.135	0.185
True distribution	150	0.113	0.113	0.148
True distribution	200	0.102	0.099	0.131

Table 3: Summary of total variation distance for samples from true distribution, based on 100 replications per sample size. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.433	0.433	0.585
A	25	0.308	0.302	0.414
A	50	0.249	0.250	0.334
A	100	0.199	0.194	0.266
A	150	0.175	0.174	0.230
A	200	0.166	0.163	0.218
B	10	0.528	0.517	0.682
B	25	0.401	0.395	0.514
B	50	0.331	0.336	0.403
B	100	0.286	0.282	0.340
B	150	0.265	0.267	0.315
B	200	0.261	0.261	0.306
C	10	0.675	0.668	0.788
C	25	0.598	0.600	0.678
C	50	0.553	0.555	0.627
C	100	0.519	0.518	0.564
C	150	0.506	0.505	0.548
C	200	0.499	0.493	0.543
D	10	0.580	0.591	0.711
D	25	0.456	0.451	0.558
D	50	0.422	0.422	0.498
D	100	0.380	0.379	0.434
D	150	0.373	0.374	0.424
D	200	0.362	0.364	0.407

Table 4: Summary of total variation distance statistics from sample simulations across convoluted distributions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.554	0.539	0.757
A	25	0.465	0.449	0.630
A	50	0.396	0.381	0.587
A	100	0.332	0.314	0.495
A	150	0.282	0.276	0.410
A	200	0.265	0.247	0.403
B	10	0.516	0.532	0.734
B	25	0.392	0.382	0.625
B	50	0.316	0.307	0.503
B	100	0.246	0.240	0.406
B	150	0.218	0.216	0.354
B	200	0.202	0.190	0.339
C	10	0.613	0.588	0.819
C	25	0.507	0.493	0.669
C	50	0.435	0.428	0.579
C	100	0.387	0.383	0.522
C	150	0.343	0.333	0.447
C	200	0.336	0.329	0.421
D	10	0.572	0.570	0.803
D	25	0.489	0.481	0.693
D	50	0.396	0.374	0.595
D	100	0.328	0.313	0.471
D	150	0.309	0.284	0.454
D	200	0.292	0.284	0.392

Table 5: Summary of total variation distance statistics from sample simulations across uniformed deconvolutions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.

model	sample_size	mean	median	upper_95
A	10	0.569	0.547	0.779
A	25	0.466	0.445	0.658
A	50	0.386	0.372	0.578
A	100	0.325	0.323	0.449
A	150	0.274	0.264	0.396
A	200	0.249	0.242	0.369
B	10	0.533	0.550	0.745
B	25	0.423	0.439	0.681
B	50	0.345	0.348	0.571
B	100	0.251	0.246	0.425
B	150	0.211	0.189	0.393
B	200	0.175	0.164	0.353
C	10	0.595	0.580	0.829
C	25	0.471	0.459	0.682
C	50	0.386	0.365	0.634
C	100	0.296	0.277	0.459
C	150	0.260	0.246	0.386
C	200	0.232	0.223	0.368
D	10	0.540	0.569	0.818
D	25	0.399	0.378	0.704
D	50	0.285	0.280	0.525
D	100	0.204	0.192	0.398
D	150	0.183	0.167	0.345
D	200	0.159	0.156	0.273

Table 6: Summary of total variation distance statistics from sample simulations across informed deconvolutions and different sample sizes, based on 100 replicas per sample siza. Total variation distance is computed relative to the true distribution.